

1. 1024 samples per transducer is recorded in the memory.

$$1024 \times 64 = 65,536$$

64 element array is
divided into $= \frac{64}{4} = 16$

The memory can be accessed as:

$$\text{Group } g \rightarrow \{4g, 4g+1, 4g+2, 4g+3\}$$

$$g = \{0, \dots, 15\}$$

2. TDM controller starts processing the 16 groups $\Rightarrow$ Always hardware reuse for all datapaths.

3. Our architecture does not calculate new geometric delays for every image pixel

Instead we have selected four fixed focal depths:

$$z_f = \{5, 9, 19, 29\} \text{ mm}$$

Controller selects one focus at a time:

$$\text{focus_sel} = \{0, 1, 2, 3\}$$

$\rightarrow$ For specific focal zones, specific delays are obtained & applied to four ADC channels.

4. In the delay LUT, the precomputed delays are indexed by $(\text{group}, \text{focus}, D_i)$ _(g)

where 'i' is the channel & index of ear transducers present inside the 16 groups.

g for each $g \in \{0, 1, 2, \dots, 15\}$



D_1, D_2, D_3, D_4

$$D_i = (D_{\text{int}})_i + \frac{(D_{\text{frac}})_i}{2^8}$$

→ These values are precomputed using a python script.

5. Delay address generation

If current ptr is $\rightarrow x[n]$

we will read previous samples according to delays.

$$\therefore \begin{matrix} x[n_i - D_i] & \& x[(n_i - D_i) + 1] \\ \{x[n]\} & & \{x[n+1]\} \end{matrix}$$

→ Then according to fractional delay value, we generate an interpolated value with delayed index & its precursors.

$$(x_{\text{interpolated}})_i = \alpha_i x[(n_i - D_i) + 1] + (1 - \alpha_i) x[n]$$

$$\alpha_i = \frac{(D_{\text{trace}})_i}{256}$$

Within a single group, $i = \{1, 2, 3, 4\}$

Hence, we get $(\alpha_{\text{interpolated}})_1, (\alpha_{\text{interpolated}})_2, \dots, (\alpha_{\text{interpolated}})_3, (\alpha_{\text{interpolated}})_4$.

$$\therefore (DAS)_{\text{group} = \text{constant}} = \sum_{i=1}^4 (\alpha_{\text{interpolated}})_i$$

★ We are not summing 64 elements into one value

→ 16 groups are processed & sequentially.

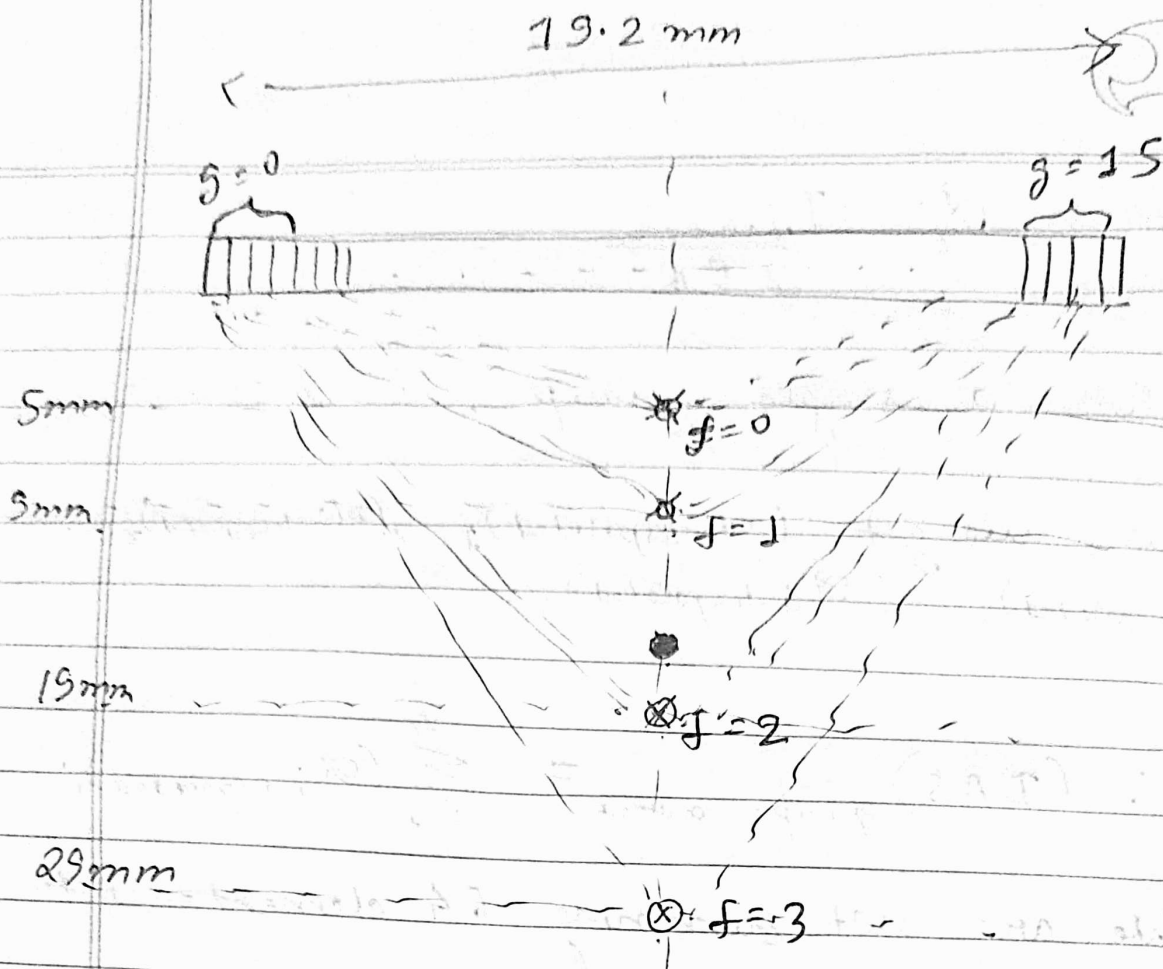
6. Process is repeated for all depth samples for a particular group & focus.

16 groups $\times$ 4 foci $\times$ 1024 samples

= 65,536 outputs per frame

Each output can be indexed as: $DAS[g, f, i]$

Note: We trace each group for a particular focal zone at once, then we move onto other focal zones.



$$\text{pitch} = 0.3 \text{ mm}$$

$$\text{total aperture} = 19.2 \text{ mm} = 64 \times 0.3 \text{ mm}$$

$$\text{element } (i^{\text{th}}) \text{ coordinate} = (i - 31.5) \times 0.3 \text{ mm}$$

$$C_{\text{sound}} = 1540 \text{ m/s}$$

$$f_s = 20 \text{ MHz}$$

$$\text{Depth resolution} = \frac{(C_{\text{sound}} \times T_s)}{2} = 0.0385 \text{ mm/sample}$$

$$\begin{aligned} \text{Acquisition depth} &= 1024 \text{ samples} \times 0.0385 \\ &= 39.4 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{ADC input format} &= 11 \text{ bit signed data} \\ &= (-512 \text{ to } +511) \end{aligned}$$